



SAFETY DATA SHEET PEARLBOND BR-10

According to the Preparation of Safety Data Sheets for Hazardous Chemicals Code of Practise, 2021.

SECTION 1: Identification: Product identifier and chemical identity

Product identifier

Product name PEARLBOND BR-10

Relevant identified uses of the substance or mixture and uses advised against

Application Di-/polyisocyanate components for the production of polyurethanes

Uses advised against Consumer spray application is not supported. Consumer applications that require heating above room temperature before or during use are not supported. Professional cleaning activities with Aprotic Polar Solvents are not supported.

Details of the supplier of the safety data sheet

Supplier Pearl Polyurethane Systems LLC
Dubai Investment Park 2
PO Box: 28605
DUBAI, UAE
UNITED ARAB EMIRATES
Tel.: +971 4 806 7900
Fax.: +971 4 8835221

Emergency telephone number

Emergency telephone +971 56 680 3698

SECTION 2: Hazard(s) identification

Classification of the substance or mixture

Physical hazards Not Classified

Health hazards Acute Tox. 4 - H332 Skin Irrit. 2 - H315 Eye Irrit. 2A - H319 Resp. Sens. 1 - H334 Skin Sens. 1 - H317
Carc. 2 - H351 STOT SE 3 - H335 STOT RE 2 - H373

Environmental hazards Not Classified

Label elements

Hazard pictograms



Signal word DANGER

Hazard statements
H332 Harmful if inhaled.
H315 Causes skin irritation.
H319 Causes serious eye irritation.
H334 May cause allergy or asthma symptoms or breathing difficulties if inhaled.
H317 May cause an allergic skin reaction.
H351 Suspected of causing cancer.
H335 May cause respiratory irritation.
H373 May cause damage to organs through prolonged or repeated exposure.



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Precautionary statements

P201 Obtain special instructions before use.
 P260 Do not breathe dust/ fume/ gas/ mist/ vapours/ spray.
 P264 Wash contaminated skin thoroughly after handling.
 P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.
 P304+P340 IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
 P308+P313 IF exposed or concerned: Get medical advice/ attention.

Supplemental label information

Contains isocyanates. May produce an allergic reaction.

Contains

Aromatic polyisocyanate, Formaldehyde, oligomeric reaction products with aniline and phosgene, 4,4'-methylenediphenyl diisocyanate, diphenylmethanediisocyanate, isomers and homologues, o-(p-isocyanatobenzyl)phenyl isocyanate

Other hazards

In case of hypersensitivity of the respiratory tract (e.g. asthmatics and those who suffer from chronic bronchitis) it is inadvisable to work with the product. Symptoms affecting the respiratory tract can also occur several hours after overexposure. Dust, vapors and aerosols are the primary risk to the respiratory tract. This mixture does not contain any substance considered as PBT or vPvB, or identified as having endocrine disrupting properties, at a concentration equal to or greater than 0.1 % by weight.

SECTION 3: Composition and information on ingredients

Mixtures

Aromatic polyisocyanate CAS number: 57288-55-8	25-50%
Classification Acute Tox. 4 - H332 Skin Irrit. 2 - H315 Eye Irrit. 2A - H319 Resp. Sens. 1 - H334 Skin Sens. 1 - H317 STOT SE 3 - H335 STOT RE 2 - H373	
Formaldehyde, oligomeric reaction products with aniline and phosgene CAS number: 32055-14-4 REACH ANNEX XVII. SCL:STOT SE 3 - H335C≥5 %Resp. Sens. 1 - H334C≥0,1 %Skin Irrit. 2 - H315C≥5 %Eye Irrit. 2 - H319C≥5 %	25-50%
Classification Acute Tox. 4 - H332 Skin Irrit. 2 - H315 Eye Irrit. 2A - H319 Resp. Sens. 1 - H334 Skin Sens. 1 - H317 Carc. 2 - H351 STOT SE 3 - H335 STOT RE 2 - H373	



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diphenylmethanediisocyanate, isomers and homologues CAS number: 9016-87-9	10-20%
Classification Skin Irrit. 2 - H315 Eye Irrit. 2A - H319 Resp. Sens. 1 - H334 Skin Sens. 1 - H317 Carc. 2 - H351 STOT SE 3 - H335 STOT RE 2 - H373	
4,4'-methylenediphenyl diisocyanate CAS number: 101-68-8 REACH ANNEX XVII. SCL: STOT SE 3 - H335C $\geq$ 5 % Resp. Sens. 1 - H334C $\geq$ 0,1 % Skin Irrit. 2 - H315C $\geq$ 5 % Eye Irrit. 2 - H319C $\geq$ 5 %	10-20%
Classification Acute Tox. 4 - H332 Skin Irrit. 2 - H315 Eye Irrit. 2A - H319 Resp. Sens. 1 - H334 Skin Sens. 1 - H317 Carc. 2 - H351 STOT SE 3 - H335 STOT RE 2 - H373	
o-(p-isocyanatobenzyl)phenyl isocyanate CAS number: 5873-54-1 Specific Concentration Limits - o-(p-isocyanatobenzyl)phenyl isocyanate: Eye Irrit. 2; H319: C $\geq$ 5 %, Resp. Sens. 1; H334: C $\geq$ 0,1 %, STOT SE 3; H335: C $\geq$ 5 %, Skin Irrit. 2; H315: C $\geq$ 5 % Lists: REACH ANNEX XVII.	5-10%
Classification Acute Tox. 4 - H332 Skin Irrit. 2 - H315 Eye Irrit. 2A - H319 Resp. Sens. 1 - H334 Skin Sens. 1 - H317 Carc. 2 - H351 STOT SE 3 - H335 STOT RE 2 - H373	



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2,2'-methylenediphenyl diisocyanate CAS number: 2536-05-2 Lists: REACH ANNEX XVII.	<1%
Classification Acute Tox. 4 - H332 Skin Irrit. 2 - H315 Eye Irrit. 2A - H319 Resp. Sens. 1 - H334 Skin Sens. 1 - H317 Carc. 2 - H351 STOT SE 3 - H335 STOT RE 2 - H373	

The full text for all hazard statements is displayed in Section 16.

SECTION 4: First aid measures

Description of first aid measures

General information	Contaminated clothing and shoes must be discarded.
Inhalation	Move affected person to fresh air and keep warm and at rest in a position comfortable for breathing. When breathing is difficult, properly trained personnel may assist affected person by administering oxygen. Get medical attention if symptoms are severe or persist.
Ingestion	Do not induce vomiting. Rinse mouth thoroughly with water. Get medical attention.
Skin Contact	In the event of contact with the skin, preferably wash with a cleanser based on polyethylene glycol or with plenty of warm water and soap. Get medical attention if any discomfort continues.
Eye contact	Remove any contact lenses and open eyelids wide apart. Continue to rinse for at least 10 minutes. Get medical attention if any discomfort continues.

Most important symptoms and effects, both acute and delayed

General information	The product irritates the respiratory tract and may trigger sensitisation of the skin and respiratory tract. Treatment of acute irritation or bronchial constriction is primarily symptomatic. Extended medical treatment may be required depending on the degree of exposure and the severity of the symptoms.
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Indication of any immediate medical attention and special treatment needed

Notes for the doctor	Treat symptomatically.
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SECTION 5: Firefighting measures

Extinguishing media

Suitable extinguishing media	Extinguish with alcohol-resistant foam, carbon dioxide, dry powder or water fog. Use fire-extinguishing media suitable for the surrounding fire.
Unsuitable extinguishing media	Do not use water jet as an extinguisher, as this will spread the fire.

Special hazards arising from the substance or mixture

Hazardous combustion products	Thermal decomposition or combustion products may include the following substances: Harmful gases or vapours.
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Advice for firefighters

Protective actions during firefighting

Evacuate area. Cool containers exposed to heat with water spray and remove them from the fire area if it can be done without risk. Cool containers exposed to flames with water until well after the fire is out. If a leak or spill has not ignited, use water spray to disperse vapours and protect men stopping the leak. Control run-off water by containing and keeping it out of sewers and watercourses.

Special protective equipment for firefighters

Wear positive-pressure self-contained breathing apparatus (SCBA) and appropriate protective clothing. Firefighter's clothing conforming to Australia/New Zealand Standards AS/NZS 4967 (for clothing) AS/NZS 1801 (for helmets), AS/NZS 4821 (for protective boots), AS/NZS 1801 (for protective gloves) will provide a basic level of protection for chemical incidents.

SECTION 6: Accidental release measures

Personal precautions, protective equipment and emergency procedures

Personal precautions

Wear protective clothing as described in Section 8 of this safety data sheet. Provide adequate ventilation. No action shall be taken without appropriate training or involving any personal risk.

Environmental precautions

Environmental precautions

Avoid discharge into drains or watercourses or onto the ground.

Methods and material for containment and cleaning up

Methods for cleaning up

Remove mechanically; cover the remainder with wet, absorbent material (e.g. sawdust, chemical binder based on calcium silicate hydrate, sand). After approx. one hour transfer to waste container and do not seal (evolution of CO₂!). Keep damp in a safe ventilated area for several days. Spill area can be decontaminated with the following recommended decontamination solution: Decontamination solution 1: 8-10% sodium carbonate and 2% of liquid soap in water Decontamination solution 2: Liquid/yellow soap (potassium soap with ~15% anionic tenside): 20ml; Water:700ml; Polyethyleneglycol (PEG 400): 350ml Decontamination solution 3: 30 % commercial laundry detergent containing monoethanolamine, 70 % water

Reference to other sections

Reference to other sections

For personal protection, see Section 8. See Section 11 for additional information on health hazards. See Section 12 for additional information on ecological hazards. For waste disposal, see Section 13.

SECTION 7: Handling and storage, including how the chemical may be safely used

Precautions for safe handling

Usage precautions

Read and follow manufacturer's recommendations. Wear protective clothing as described in Section 8 of this safety data sheet. Keep away from food, drink and animal feeding stuffs. Provide adequate ventilation.

Advice on general occupational hygiene

Wash promptly if skin becomes contaminated. Take off contaminated clothing and wash before reuse. Wash contaminated clothing before reuse.

Conditions for safe storage, including any incompatibilities

Storage precautions

Keep away from food, drink and animal feeding stuffs. Keep only in the original container. Keep container tightly closed, in a cool, well ventilated place.

Specific end use(s)

Specific end use(s)

The identified uses for this product are detailed in Section 1.



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SECTION 8: Exposure controls and personal protection

Control parameters

Occupational exposure limits

Formaldehyde, oligomeric reaction products with aniline and phosgene

Long-term exposure limit (8-hour TWA): 0.02 mg/m³

Short-term exposure limit (15-minute): 0.07 mg/m³

Carc. 2, Sen

4,4'-methylenediphenyl diisocyanate

Long-term exposure limit (8-hour TWA): 0.02 mg/m³

Short-term exposure limit (15-minute): 0.07 mg/m³

Carc. 2, Sen

Carc. 2 = Suspected human carcinogen.

Sen = Respiratory and/or skin sensitiser.

Exposure controls

Protective equipment



Appropriate engineering controls Provide adequate ventilation.

Eye/face protection Wear eye protection. Standard EN 166 - Personal eye-protection.

Hand protection Avoid contact with skin. It must comply with the EN374 standard. It is recommended that gloves are made of the following material: Nitrile rubber. Butyl rubber. Neoprene. Viton rubber (fluoro rubber). Polychloroprene - CR: thickness $\geq 0,5\text{mm}$; breakthrough time $\geq 480\text{min}$. Nitrile rubber - NBR: thickness $\geq 0,35\text{mm}$; breakthrough time $\geq 480\text{min}$. Butyl rubber - IIR: thickness $\geq 0,5\text{mm}$; breakthrough time $\geq 480\text{min}$. Fluorinated rubber - FKM: thickness $\geq 0,4\text{mm}$; breakthrough time $\geq 480\text{min}$.

Hygiene measures Wash hands thoroughly after handling. Do not eat, drink or smoke when using this product. Wash contaminated clothing before reuse.

Respiratory protection Provide adequate ventilation. If ventilation is inadequate, suitable respiratory protection must be worn. Combination filter, type A2/P2.

Environmental exposure controls Keep container tightly sealed when not in use. Avoid release to the environment.

SECTION 9: Physical and chemical properties

Information on basic physical and chemical properties

Appearance Liquid. at 20 °C at 1.013 hPa

Colour Brown.

Odour Slight.

Odour threshold No information available.

pH Not applicable.

Melting point No information available.

Initial boiling point and range $> 300^{\circ}\text{C}$ (DIN 53171)

Flash point 213°C (DIN EN 22719)



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Evaporation rate	No information available.
Evaporation factor	No information available.
Flammability (solid, gas)	No information available.
Flammability Limit - Lower(%)	No information available.
Vapour pressure	For products with a very low vapor pressure, the apparent vapor pressure may exceed the vapor pressure of the pure product due to conditions of manufacturing, storage or transportation, e.g. by solved gases like nitrogen or carbon dioxide 6 hPa at 20 °CEG A4 18 hPa at 50 °CEG A4 21 hPa at 55 °CEG A4
Vapour density	No information available.
Relative density	No information available.
Density or relative density	1,216 g/cm ³ @ 20 °C DIN 51757
Bulk density	No information available.
Solubility(ies)	Immiscible with water.@ 15 °C
Partition coefficient	No information available.
Auto-ignition temperature	Not applicable.
Decomposition Temperature	No information available.
Viscosity	2200 mPa.s at 25 °C
Explosive properties	No information available.
Oxidising properties	No information available.
Particle characteristics	Not applicable.
Pour Point (°C)	-19 °C (ISO 3016)
Ignition temperature	> 500 °C (DIN 51794)
Other information	No information required.

SECTION 10: Stability and reactivity

Reactivity	There are no known reactivity hazards associated with this product.
Stability	Stable at normal ambient temperatures and when used as recommended. Stable under the prescribed storage conditions.
Possibility of hazardous reactions	No potentially hazardous reactions known.
Conditions to avoid	There are no known conditions that are likely to result in a hazardous situation.
Materials to avoid	No specific material or group of materials is likely to react with the product to produce a hazardous situation.
Hazardous decomposition products	Does not decompose when used and stored as recommended. Thermal decomposition or combustion products may include the following substances: Harmful gases or vapours.



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SECTION 11: Toxicological information

Information on toxicological effects

Acute toxicity - oral

Notes (oral LD₅₀) Based on available data the classification criteria are not met.

Acute toxicity - dermal

Notes (dermal LD₅₀) Based on available data the classification criteria are not met.

Acute toxicity - inhalation

Notes (inhalation LC₅₀) Acute Tox. 4 - H332 Harmful if inhaled.

ATE inhalation (vapours mg/l) 24.94

ATE inhalation (dusts/mists mg/l) 3.13

Skin corrosion/irritation

Skin corrosion/irritation Causes skin irritation.

Serious eye damage/irritation

Serious eye damage/irritation Causes serious eye irritation.

Respiratory sensitisation

Respiratory sensitisation There is evidence that the product can cause respiratory hypersensitivity.

Skin sensitisation

Skin sensitisation May cause skin sensitisation or allergic reactions in sensitive individuals.

Germ cell mutagenicity

Genotoxicity - in vitro Based on available data the classification criteria are not met.

Carcinogenicity

Carcinogenicity Suspected of causing cancer.

Reproductive toxicity

Reproductive toxicity - fertility Based on available data the classification criteria are not met.

Reproductive toxicity - development

Based on available data the classification criteria are not met.

Specific target organ toxicity - single exposure

STOT - single exposure STOT SE 3 - H335 May cause respiratory irritation.

Target organs

Respiratory system, lungs

Specific target organ toxicity - repeated exposure

STOT - repeated exposure STOT RE 2 - H373 May cause damage to organs through prolonged or repeated exposure.

Aspiration hazard

Aspiration hazard Based on available data the classification criteria are not met.



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Information on other hazards

Toxicological information on ingredients. This mixture does not contain any substance considered as having endocrine disrupting properties

Aromatic polyisocyanate

Acute toxicity - oral

Notes (oral LD₅₀) LD₅₀ >2000 mg/kg, Oral, Rat Based on the test results of similar product.

Acute toxicity - dermal

Notes (dermal LD₅₀) LD₅₀ >9400 mg/kg, Dermal, Rabbit (OECD Test Guideline 402)
Based on the test results of similar product.

Acute toxicity - inhalation

ATE inhalation (dusts/mists mg/l) 1.5

Respiratory sensitisation

Respiratory sensitisation Local Lymph Node Assay (LLNA) Mouse: Sensitising. OECD Guideline 429
Based on the test results of similar product.

Germ cell mutagenicity

Genotoxicity - in vitro Ames test: Negative with metabolic activation., Negative without metabolic activation. (OECD Guideline 471) Based on the test results of similar product.

Genotoxicity - in vivo Micronucleus assay: Negative. Inhalation (OECD Guideline 474) Based on the test results of similar product.

Reproductive toxicity

Reproductive toxicity - development Teratogenicity: - NOAEL: 12 mg/m³, Inhalation, Rat Maternal toxicity: - NOAEL: 4 mg/m³, Inhalation, Rat Developmental toxicity: - NOAEL: 4 mg/m³, Inhalation, Rat 20 day Developmental toxicity: - NOAEL: 4 mg/m³, Note: Did not show teratogenic effects in animal experiments. Based on the test results of similar product.

Specific target organ toxicity - repeated exposure

STOT - repeated exposure NOAEL: 0,2 mg/m³ LOAEL (Lowest observable adverse effect level): 1 mg/m³ Application Route: Inhalative Species: rat, male/female Dose Levels: 0 - 0,2 - 1 - 6 mg/m³ Exposure duration: 2 a Frequency of treatment: 6 hours a day, 5 days a week Target Organs: Lungs, Nasal inner lining Test substance: as aerosol Method: OECD Test Guideline 453 Findings: Irritation to nasal cavity and to lungs. Studies of a comparable product.

Formaldehyde, oligomeric reaction products with aniline and phosgene

Acute toxicity - oral

Notes (oral LD₅₀) LD₅₀ >10000 mg/kg, Oral, Rat (OECD Test Guideline 401)
Based on the test results of similar product.

Acute toxicity - dermal

Notes (dermal LD₅₀) LD₅₀ >9.400 mg/kg, Dermal, Rabbit (OECD Test Guideline 402)
Based on the test results of similar product.

Acute toxicity - inhalation

Notes (inhalation LC₅₀) LC50 0.31 mg/l, Inhalation, Rat (OECD Test Guideline 403)

ATE inhalation (vapours mg/l) 11.0



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ATE inhalation (dusts/mists mg/l)	1.5
<i>Skin corrosion/irritation</i>	
Skin corrosion/irritation	Slightly irritating. Rabbit (OECD 404) Based on the test results of similar product.
<i>Serious eye damage/irritation</i>	
Serious eye damage/irritation	Not irritating. Rabbit (OECD 405) Based on the test results of similar product.
<i>Respiratory sensitisation</i>	
Respiratory sensitisation	Rat: Sensitising. May cause allergy or asthma symptoms or breathing difficulties if inhaled. Based on the test results of similar product.
<i>Skin sensitisation</i>	
Skin sensitisation	Guinea pig maximization test (GPMT) - Guinea pig: Not sensitising. (OECD 406) Based on the test results of similar product. Local Lymph Node Assay (LLNA) - Mouse: Sensitising. OECD Guideline 429 Based on the test results of similar product.
<i>Germ cell mutagenicity</i>	
Genotoxicity - in vitro	Ames test: Negative with metabolic activation., Negative without metabolic activation. Salmonella typhimurium (OECD Guideline 471) Based on the test results of similar product.
Genotoxicity - in vivo	Micronucleus assay: Negative. Inhalation Rat (OECD Guideline 474) Based on the test results of similar product.
<i>Carcinogenicity</i>	
Carcinogenicity	0-0.2-1-6 mg/m ³ , Inhalation, Rat (OECD Guideline 453) Based on the test results of similar product.
<i>Reproductive toxicity</i>	
Reproductive toxicity - development	Teratogenicity: - NOAEL: 12 mg/m ³ , Inhalation, Rat Maternal toxicity: - NOAEL: 4 mg/m ³ , Inhalation, Rat Developmental toxicity: - NOAEL: 4 mg/m ³ , Inhalation, Rat Dose: 0-1-4-12, , 20 day OECD 414 Note: Did not show teratogenic effects in animal experiments. Based on the test results of similar product.
<i>Specific target organ toxicity - repeated exposure</i>	
STOT - repeated exposure	NOAEL: 0,2 mg/m ³ LOAEL (Lowest observable adverse effect level): 1 mg/m ³ Application Route: Inhalative Species: rat, male/female Dose Levels: 0 - 0,2 - 1 - 6 mg/m ³ Exposure duration: 2 a Frequency of treatment: 6 hours a day, 5 days a week Target Organs: Lungs, Nasal inner lining Test substance: as aerosol Method: OECD Test Guideline 453 Findings: Irritation to nasal cavity and to lungs. Studies of a comparable product.
<i>diphenylmethanediisocyanate, isomers and homologues</i>	
<i>Acute toxicity - oral</i>	
Notes (oral LD₅₀)	LD ₅₀ >2000 mg/kg, Oral, Rat (OECD 401) Based on the test results of similar product.
<i>Acute toxicity - dermal</i>	
Notes (dermal LD₅₀)	LD ₅₀ >9400 mg/kg, Dermal, Rabbit (OECD 402)
<i>Acute toxicity - inhalation</i>	



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Notes (inhalation LC₅₀)	LC50 0.31 mg/l, Inhalation, Rat (OECD Test Guideline 403)
<i>Skin corrosion/irritation</i>	
Skin corrosion/irritation	Slightly irritating. Rabbit (OECD Test Guideline 404)
<i>Serious eye damage/irritation</i>	
Serious eye damage/irritation	Not irritating. Rabbit (OECD 405) Based on the test results of similar product.
<i>Skin sensitisation</i>	
Skin sensitisation	Guinea pig maximization test (GPMT) Negative. Not sensitising. (OECD Test Guideline 406) Based on the test results of similar product.
<i>Germ cell mutagenicity</i>	
Genotoxicity - in vitro	Ames test: Negative. Salmonella typhimurium Negative without metabolic activation. Negative with metabolic activation. (OECD Guideline 471)
Genotoxicity - in vivo	Micronucleus assay: Negative. (OECD Guideline 474) Based on the test results of similar product.
<i>Carcinogenicity</i>	
Carcinogenicity	0-0,2-1-6 mg/m ³ , Inhalation, Rat Frequency of treatment: 6 hours/day, 5 days/week (OECD Guideline 453)
<i>Reproductive toxicity</i>	
Reproductive toxicity - development	Teratogenicity: - NOAEL: 12 mg/m ³ , , Rat, Female. Maternal toxicity: - NOAEL: 4 mg/m ³ , , Rat, Female. Developmental toxicity: - NOAEL: 4 mg/m ³ , , Rat, Female. OECD 414 Developmental toxicity: - NOAEL: 4 mg/m ³ , Note: Did not show teratogenic effects in animal experiments.

Specific target organ toxicity - repeated exposure

STOT - repeated exposure	NOAEL: 0,2 mg/m ³ LOAEL (Lowest observable adverse effect level): 1 mg/m ³ Application Route: Inhalative Species: rat, male/female Dose Levels: 0 - 0,2 - 1 - 6 mg/m ³ Exposure duration: 2 a Frequency of treatment: 6 hours a day, 5 days a week Target Organs: Lungs, Nasal inner lining Test substance: as aerosol Method: OECD Test Guideline 453 Findings: Irritation to nasal cavity and to lungs. Studies of a comparable product.
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4,4'-methylenediphenyl diisocyanate

Acute toxicity - oral

Notes (oral LD₅₀)	LD ₅₀ >2000 mg/kg, Oral, Rat (OECD Test Guideline 401) Based on the test results of similar product.
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Acute toxicity - dermal

Notes (dermal LD₅₀)	LD ₅₀ >9.400 mg/kg, Dermal, Rabbit (OECD Test Guideline 402) Based on the test results of similar product.
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Acute toxicity - inhalation

Notes (inhalation LC₅₀)	LC50 0.368 mg/l, Inhalation, Rat (OECD Test Guideline 403)
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ATE inhalation (vapours mg/l)	11.0
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ATE inhalation (dusts/mists mg/l)	1.5
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Skin corrosion/irritation

Skin corrosion/irritation Causes skin irritation. Rabbit (OECD 404) Based on the test results of similar product.

Serious eye damage/irritation

Serious eye damage/irritation Not irritating. Rabbit (OECD 405) Based on the test results of similar product.

Respiratory sensitisation

Respiratory sensitisation Guinea pig: Sensitising.

Skin sensitisation

Skin sensitisation Buehler test - Guinea pig: Not sensitising. (OECD 406) Local Lymph Node Assay (LLNA) - Mouse: Sensitising. OECD Guideline 429

Germ cell mutagenicity

Genotoxicity - in vitro Ames test: Negative with metabolic activation., Negative without metabolic activation. (OECD 471) Based on the test results of similar product.

Genotoxicity - in vivo Micronucleus assay: Negative. (OECD Guideline 474) comet assay: Negative. Dose: 2-5-11 mg/m³, , Rat OECD 489

Carcinogenicity

Carcinogenicity Dose level: 0-0.2-1-6 mg/m³, Inhalation, Rat (OECD Guideline 453) Based on the test results of similar product.

Reproductive toxicity

Reproductive toxicity - development Teratogenicity: - NOAEL: 12 mg/m³, Inhalation, Rat Maternal toxicity: - NOAEL: 4 mg/m³, Inhalation, Rat Developmental toxicity: - NOAEL: 4 mg/m³, Inhalation, Rat Dose level: 0-1-4-12 mg/m³ 20 day OECD 414 Developmental toxicity: - NOAEL: 4 mg/m³, Note: Did not show teratogenic effects in animal experiments. Based on the test results of similar product.

Specific target organ toxicity - repeated exposure

STOT - repeated exposure NOAEL: 0,2 mg/m³ LOAEL (Lowest observable adverse effect level): 1 mg/m³ Application Route: Inhalative Species: rat, male/female Dose Levels: 0 - 0,2 - 1 - 6 mg/m³ Exposure duration: 2 a Frequency of treatment: 6 hours a day, 5 days a week Target Organs: Lungs, Nasal inner lining Test substance: as aerosol Method: OECD Test Guideline 453 Findings: Irritation to nasal cavity and to lungs. Studies of a comparable product.

o-(p-isocyanatobenzyl)phenyl isocyanate

Acute toxicity - inhalation

ATE inhalation (vapours mg/l) 11.0

ATE inhalation (dusts/mists mg/l) 1.5

SECTION 12: Ecological information

Ecotoxicity Not regarded as dangerous for the environment. However, large or frequent spills may have hazardous effects on the environment.

Toxicity Based on available data the classification criteria are not met.

Ecological information on ingredients.



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Aromatic polyisocyanate

Acute aquatic toxicity

Acute toxicity - fish	LC ₅₀ , 96 hour: >1000 mg/l, Danio rerio (zebra fish) (OECD 203) Similar substance.
Acute toxicity - aquatic invertebrates	EC ₅₀ , 24 hour: >1000 mg/l, Daphnia magna (OECD 202) Based on the test results of similar product. NOEC, 21 day: >10 mg/l, Daphnia magna (OECD 202) Based on the test results of similar product.
Acute toxicity - aquatic plants	ErC ₅₀ , 72 hour: >1640 mg/l, Scenedesmus subspicatus (OECD 201) Based on the test results of similar product.
Acute toxicity - microorganisms	EC ₅₀ , 3 hour: >100 mg/l, Activated sludge (OECD 209) Based on the test results of similar product.

Formaldehyde, oligomeric reaction products with aniline and phosgene

Acute aquatic toxicity

Acute toxicity - fish	LC ₅₀ , 96 hour: >1000 mg/l, Danio rerio (zebra fish) (OECD 203) Similar substance.
Acute toxicity - aquatic invertebrates	EC ₅₀ , 24 hour: >1000 mg/l, Daphnia magna (OECD 202) Based on the test results of similar product. NOEC, 21 day: >10 mg/l, Daphnia magna (OECD 202) Based on the test results of similar product.
Acute toxicity - aquatic plants	ErC ₅₀ , 72 hour: >1640 mg/l, Scenedesmus subspicatus (OECD 201) Based on the test results of similar product.
Acute toxicity - microorganisms	EC ₅₀ , 3 hour: >100 mg/l, Activated sludge (OECD 209) Based on the test results of similar product.
Acute toxicity - terrestrial	NOEC, 14 day: >1000 mg/kg, Eisenia Fetida (Earthworm) OECD 207 NOEC, 14 day: >1000 mg/kg, Avena sativa OECD 208 Based on the test results of similar product.

diphenylmethanediisocyanate, isomers and homologues

Acute aquatic toxicity

Acute toxicity - fish	LC ₅₀ , 96 hour: >1000 mg/l, Danio rerio (zebra fish) (OECD 203)
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According to the Preparation of Safety Data Sheets for Hazardous Chemicals Code of Practice, 2021.

Acute toxicity - aquatic invertebrates	EC ₅₀ , 24 hour: >1000 mg/l, Daphnia magna (OECD 202) NOEC, 21 day: >10 mg/l, Daphnia magna (OECD 211)
Acute toxicity - aquatic plants	ErC ₅₀ , 72 hour: >1640 mg/l, Scenedesmus subspicatus (OECD 201)
Acute toxicity - microorganisms	EC ₅₀ , 3 hour: >100 mg/l, Activated sludge (OECD 209)
Acute toxicity - terrestrial	NOEC, 14 day: >1000 mg/kg, Eisenia Fetida (Earthworm) OECD 207 NOEC, 14 day: >1000 mg/kg, Avena sativa OECD 208

4,4'-methylenediphenyl diisocyanate

Acute aquatic toxicity

Acute toxicity - fish	LC ₅₀ , 96 hour: >1000 mg/l, Danio rerio (zebra fish) Based on the test results of similar product.
Acute toxicity - aquatic invertebrates	EC ₅₀ , 24 hour: >1000 mg/l, Daphnia magna (OECD 202) Based on the test results of similar product. NOEC, 21 day: >10 mg/l, Daphnia magna (OECD 202) Based on the test results of similar product.
Acute toxicity - aquatic plants	ErC ₅₀ , 72 hour: >1640 mg/l, Scenedesmus subspicatus (OECD 201) Based on the test results of similar product.
Acute toxicity - microorganisms	EC ₅₀ , 3 hour: >100 mg/l, Activated sludge (OECD 209) Based on the test results of similar product.
Acute toxicity - terrestrial	NOEC, 14 day: >1000 mg/kg, Eisenia Fetida (Earthworm) OECD 207 Based on the test results of similar product. NOEC, 14 day: >1000 mg/kg, Avena sativa OECD 208 Based on the test results of similar product.

Persistence and degradability

Persistence and degradability The degradability of the product is not known.

Ecological information on ingredients.

Aromatic polyisocyanate

Persistence and degradability	% 0 28 day Not inherently biodegradable. OECD 302 C Based on the test results of similar product.
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Formaldehyde, oligomeric reaction products with aniline and phosgene

Persistence and degradability 0 % 28 day OECD 302 C Not readily biodegradable.

diphenylmethanediisocyanate, isomers and homologues

Persistence and degradability 0 % 28 day OECD 302 C Not readily biodegradable.

4,4'-methylenediphenyl diisocyanate

Persistence and degradability 0 % 28 day OECD 302 C Not readily biodegradable.

Bioaccumulative potential

Bioaccumulative Potential No data available on bioaccumulation.

Partition coefficient No information available.

Ecological information on ingredients.

Formaldehyde, oligomeric reaction products with aniline and phosgene

Bioaccumulative Potential BCF: 200, Cyprinus carpio (Common carp)
: 0.0008 mg/l 28 gün, Cyprinus carpio (Common carp)

Partition coefficient log Pow: 4.51 (72 °F / 22 °C)

diphenylmethanediisocyanate, isomers and homologues

Bioaccumulative Potential BCF: 92, Cyprinus carpio (Common carp) 28 day (OECD 305) Based on the test results of similar product.
BCF: 200, Cyprinus carpio (Common carp) (OECD 305) Accumulation in organisms is not to be expected.

4,4'-methylenediphenyl diisocyanate

Bioaccumulative Potential BCF: 200, Cyprinus carpio (Common carp)
: 0.0008 mg/l 28 gün, (OECD 305) Accumulation in organisms is not to be expected.

o-(p-isocyanatobenzyl)phenyl isocyanate

Partition coefficient log Pow: 5.22 (77 °F / 25 °C)

Mobility in soil

Mobility No data available.

Other adverse effects

Other adverse effects Isocyanate reacts with water at the interface forming CO₂ and a solid insoluble product with high melting point (polyurea). This reaction is accelerated by surfactants (e.g. detergents) or by watersoluble solvents. Previous experience shows that polyurea is inert and non-degradable.

SECTION 13: Disposal considerations

Waste treatment methods

General information The generation of waste should be minimised or avoided wherever possible. Reuse or recycle products wherever possible. This material and its container must be disposed of in a safe way. When handling waste, the safety precautions applying to handling of the product should be considered. Empty containers or liners may retain some product residues and hence be potentially hazardous.



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According to the Preparation of Safety Data Sheets for Hazardous Chemicals Code of Practice, 2021.

Disposal methods Dispose of waste to licensed waste disposal site in accordance with the requirements of the local Waste Disposal Authority.

SECTION 14: Transport information

General The product is not covered by international regulations on the transport of dangerous goods (IMDG, IATA, ADG).

UN number or ID number

Not applicable.

UN proper shipping name

Not applicable.

Transport hazard class(es)

No transport warning sign required.

Packing group

Not applicable.

Environmental hazards

Environmentally hazardous substance/marine pollutant

No.

Special precautions for user

Not applicable.

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code Not applicable.

SECTION 15: Regulatory information

Safety, health and environmental regulations/legislation specific for the substance or mixture

National regulations Preparation of Safety Data Sheets for Hazardous Chemicals - Code of Practice
GHS (Globally Harmonised System)

WGK 1 slightly water endangering

AIIC On the inventory, or in compliance with the inventory.



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According to the Preparation of Safety Data Sheets for Hazardous Chemicals Code of Practice, 2021.

SECTION 16: Any other relevant information

Abbreviations and acronyms used in the safety data sheet	ADG: Australian dangerous goods code IATA: International air transport association. ICAO: Technical instructions for the safe transport of dangerous goods by air. IMDG: International maritime dangerous goods. CAS: Chemical abstracts service. ATE: Acute toxicity estimate. LC₅₀: Lethal concentration to 50 % of a test population. LD₅₀: Lethal dose to 50% of a test population (median lethal dose). EC₅₀: 50% of maximal effective concentration. PBT: Persistent, bioaccumulative and toxic substance. vPvB: Very persistent and very bioaccumulative.
Classification abbreviations and acronyms	Acute Tox. = Acute toxicity Carc. = Carcinogenicity Eye Irrit. = Eye irritation Resp. Sens. = Respiratory sensitisation Skin Irrit. = Skin irritation Skin Sens. = Skin sensitisation STOT RE = Specific target organ toxicity-repeated exposure STOT SE = Specific target organ toxicity-single exposure
Key literature references and sources for data	This SDS is prepared based on the information and documents received from product owner. CRAD or/and SDS author shall not be responsible for incorrect prepared of SDS and pecuniary loss or intangible damages because of deficient or wrong information and documents which comes from product owner.
Revision comments	This is the first issue.
Issued by	Büşra TARAKCI / CRAD gbf@crad.com.tr Tel+90 216 3354600
Revision date	2/01/2025
Revision	1.0
Supersedes date	2/01/2025
SDS No.	15854
Hazard statements in full	H315 Causes skin irritation. H317 May cause an allergic skin reaction. H319 Causes serious eye irritation. H332 Harmful if inhaled. H334 May cause allergy or asthma symptoms or breathing difficulties if inhaled. H335 May cause respiratory irritation. H351 Suspected of causing cancer. H351 Suspected of causing cancer if swallowed. H373 May cause damage to organs through prolonged or repeated exposure. H373 May cause damage to organs through prolonged or repeated exposure if swallowed or if inhaled.

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